

Data Cleaning & Preprocessing

Internship Application Data Using Python and Pandas

Project Objective	Ensure data accuracy by cleaning and transforming raw internship application data.
Technology	Python, Pandas, NumPy, Jupyter Notebook
Dataset Type	Internship application data: applicant demographics, department, skills, application status, experience
Final Output	Cleaned CSV dataset prepared for analysis, reporting, and visualization

1. Project Overview

This project demonstrates a practical data cleaning and preprocessing workflow for internship application records. The raw data is inspected for missing values, duplicate records, inconsistent text, unrealistic numerical values, and potential outliers. The cleaned data is then standardized and exported as a CSV file for further analysis.

2. Cleaning Tasks Completed

Requirement	Action Performed
Missing values	Checked null values and replaced missing Age values using the median.
Duplicate records	Checked duplicated rows and removed duplicates.
Outliers	Checked unrealistic Age values and demonstrated IQR-based outlier detection.
Standardization	Standardized Gender and Department values using string formatting and replacement.
Data structure	Inspected rows, columns, data types, and dataset dimensions.
Automation	Used Pandas and NumPy so the cleaning process is reproducible.
Export	Saved the cleaned dataset as internship_applications_cleaned.csv.

3. Python Implementation

```
import pandas as pd
import numpy as np
```

```
# Load or create the internship application data
df = pd.DataFrame(data)
```

```
# Inspect the dataset
print(df.shape)
print(df.info())
print(df.isnull().sum())
```

Handle Missing Values

```
# Fill missing Age values with the median
df["Age"] = df["Age"].fillna(df["Age"].median())
```

```
# Check again
print(df.isnull().sum())
```

Remove Duplicate Records

```
print("Duplicates before:", df.duplicated().sum())
df = df.drop_duplicates()
print("Duplicates after:", df.duplicated().sum())
```

Standardize Text Fields

```
df["Gender"] = df["Gender"].str.strip().str.title()
df["Department"] = df["Department"].str.strip().str.title()

# Standardize selected department labels
df["Department"] = df["Department"].replace({"It": "IT", "Hr": "HR"})
```

4. Outlier Detection and Treatment

An unrealistic Age value such as 999 is treated as an invalid observation rather than a legitimate applicant age. The value is converted to missing and then replaced with the median. For other numerical variables, the Interquartile Range (IQR) method can be used to identify unusually high or low values.

```
# Handle unrealistic Age values
df.loc[(df["Age"] < 18) | (df["Age"] > 60), "Age"] = np.nan
df["Age"] = df["Age"].fillna(df["Age"].median())

# IQR method for Experience_Months
Q1 = df["Experience_Months"].quantile(0.25)
Q3 = df["Experience_Months"].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR

outliers = df[(df["Experience_Months"] < lower) |
               (df["Experience_Months"] > upper)]
print(outliers)
```

5. Final Validation

```
print("Missing values:")
print(df.isnull().sum())

print("Duplicate rows:")
print(df.duplicated().sum())

print("Age range:", df["Age"].min(), "to", df["Age"].max())
```

6. Export Cleaned Dataset

```
df.to_csv("internship_applications_cleaned.csv", index=False)
print("Cleaned dataset saved successfully!")
```

Expected Project Structure

File	Purpose
Data_Cleaning_and_Preprocessing_Internship.ipynb	Complete Python/Jupyter workflow
internship_applications_cleaned.csv	Final cleaned dataset
README.md	GitHub project description and documentation
Data_Cleaning_Preprocessing_Internship_Report.pdf	Project report for submission/portfolio

7. Submission Summary

The cleaned internship application dataset is suitable for downstream analysis and visualization. The workflow demonstrates the required use of Python/Pandas for data quality checks, missing-value handling, duplicate removal, outlier detection, standardization, and automated export.